Thinking, Decisions, Intelligence and Language

How people solve problems and make judgements, the predictable shortcuts that lead them astray, what IQ does and doesn't measure, and how language shapes thought.

Isagi · November 2026 · Spine: Ciccarelli & White, *Psychology* (4th ed.), Ch. 7 — rewritten, updated and applied, with heuristics and biases expanded

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Management is decision-making under uncertainty with too little time. Psychology's most useful contribution here is not a theory of how people should decide but a catalogue of how they actually do: fast shortcuts that usually work and fail in predictable ways. The textbook treats this briefly; this part expands it, because heuristics and biases are where psychology meets pricing, negotiation, forecasting and hiring. The second half covers intelligence — what IQ tests measure, how well they predict work performance, and which popular alternatives hold up.

1 The building blocks of thought

Cognition is the mental activity of thinking, understanding, remembering and communicating. Two of its building blocks:

- **Mental images** — picture-like representations. We manipulate them much as we would real objects: in classic experiments, the time people take to mentally rotate a shape rises with the angle of rotation.
- **Concepts** — mental categories that group objects, events or ideas. **Formal concepts** have strict rules (a triangle has three sides). **Natural concepts** form through experience and have fuzzy edges. We judge category members against a **prototype**, the most typical example. Ask someone to name a "premium car" and you'll hear a few prototypes; a brand that sits close to the prototype gets the category's associations for free.

2 Problem solving

There are three broad ways to solve a problem. **Trial and error** works for simple problems with few options. An **algorithm** is a step-by-step procedure that guarantees a solution if followed — a checklist, a formula, a pricing calculator — but can be slow. A **heuristic** is a rule of thumb that is fast and usually works, but can fail: "price at the competitor's level", "work backwards from the goal", "break it into smaller problems".

Three barriers block good solutions:

- **Functional fixedness** — seeing objects only in their usual role. In Karl Duncker's candle problem, people given a candle, matches and a box of drawing pins struggle to fix the candle to a wall. The answer is to empty the box and pin it up as a shelf. Seeing the box only as a container is the block. Business version: seeing a distributor only as a logistics channel, not as a source of market intelligence.
- **Mental set** — sticking with a method that worked before, even when a simpler one exists.
- **Confirmation bias** — seeking and favouring evidence that supports what you already believe, and ignoring what doesn't. It is the most damaging bias in reviews and investigations: once a manager decides a rep is weak, every missed call confirms it and every good month is "luck".

3 Heuristics and biases

From the 1970s Daniel Kahneman and Amos Tversky showed that human judgement departs from logic in systematic, predictable ways. Kahneman later popularised a two-system description. **System 1** is fast, automatic and intuitive; **System 2** is slow, effortful and deliberate. Most judgements are made by System 1, and System 2 is lazy about checking them. The main shortcuts:

Availability

We judge how likely something is by how easily examples come to mind. Vivid, recent or emotional events feel more common than they are. After one dealer defaults loudly, a sales head may tighten credit for everyone, though the default rate hasn't changed.

Representativeness and base rates

We judge whether something belongs to a category by how much it resembles our prototype, and ignore how common the category actually is (the **base rate**). A candidate who "looks like" your star salesperson — confident, talkative, a former athlete — feels like a likely star, even if only a small share of such candidates ever become stars.

WORKED EXAMPLE 1 · BASE-RATE NEGLECT IN HIRING

Of 100 sales hires, historically 10 become top performers. Your interviewers are fairly good: they rate 80% of future top performers as "star material", but also rate 20% of the others that way. A candidate is rated "star material". What's the chance they become a top performer?

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Future stars rated "star":    80% of 10 =  8
Others rated "star":          20% of 90 = 18
Total rated "star":                       26

P(top performer | rated star) = 8 / 26 ≈ 31%
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Most people guess around 80%. The true figure is about 31%, because ordinary performers vastly outnumber stars. This is the same arithmetic as the reverse-inference trap in Part 1. When the base rate is low, even a good test produces mostly false positives.

Anchoring

An initial number pulls later estimates toward it, even when it is obviously irrelevant. In a 1974 study, Tversky and Kahneman spun a wheel rigged to stop at 10 or 65, then asked people what percentage of UN member countries were African. Those who saw 10 gave a median

estimate of 25%; those who saw 65, 45%. A random number moved the answer by 20 points. In negotiation, the first price on the table anchors the range — which is why the side that opens with a credible, ambitious number usually ends better off, and why MRPs are printed well above the street price.

Framing

The same choice described differently leads to different decisions. In Tversky and Kahneman's 1981 "disease problem", people chose between programmes to fight an outbreak expected to kill 600. Described in terms of lives saved ("200 people will be saved"), 72% chose the certain option. Described in terms of lives lost ("400 people will die"), only 22% did. The outcomes were identical.

Loss aversion

Losses hurt more than equal gains please — by roughly a factor of two in many studies. Most people refuse a coin flip to win ₹1,500 or lose ₹1,000, despite its positive expected value, and typically want a potential gain about twice the potential loss before they accept. This is why "don't lose your 5% loyalty bonus" can motivate more than "earn a 5% bonus", and why customers stick with what they have (the **status quo bias**).

Sunk cost and overconfidence

The **sunk-cost fallacy** is continuing to invest in something because of what you've already spent — the failing product launch kept alive "because we've already put ₹2 crore into it". Only future costs and benefits should count. **Overconfidence** is the near-universal tendency to be more sure of our judgements than accuracy justifies. When people say they are "90% sure" of a range, the truth often falls inside it far less than 90% of the time. Forecasts need wider ranges than feel natural.

DEBIASING A DECISION: FIVE HABITS

1. Ask for the base rate first: "How often does this usually happen?"
2. Reframe: state the choice as a gain and as a loss and see if your answer changes.
3. Make your own estimate before you see anyone else's number.
4. Ignore sunk costs: "If we were starting today, would we invest?"
5. Assign someone to argue the other side, and run a "pre-mortem": imagine it failed and list why.

4 Creativity

Convergent thinking narrows to a single correct answer. **Divergent thinking** generates many possible answers — the core of creativity. A practical finding: group brainstorming, in which people shout out ideas together, reliably produces *fewer* and no better ideas than the same number of people working alone and then pooling their lists. The causes are people waiting their turn and forgetting their idea (**production blocking**), fear of judgement, and coasting. The better method: individuals write ideas silently first, then the group combines and builds on them.

5 Intelligence

A working definition: **intelligence** is the ability to learn from experience, solve problems, and adapt to new situations. The main theories:

- **Spearman's g** (1904): people who do well on one kind of mental test tend to do well on others. He proposed a general ability, *g*, plus specific abilities. This pattern is one of the most replicated findings in psychology.
- **Gardner's multiple intelligences** (1983): eight or so separate intelligences — linguistic, logical-mathematical, spatial, musical, bodily-kinaesthetic, interpersonal, intrapersonal, naturalist. Popular in schools, but the proposed intelligences correlate with one another and with g, and there is little evidence they are independent.

- **Sternberg's triarchic theory**: **analytical** (the kind IQ tests measure), **creative**, and **practical** intelligence ("street smarts"). Useful as a reminder that test smarts aren't the whole story.

Measuring it

Alfred Binet built the first practical test in Paris in 1905 to identify children who needed extra help at school. The modern standard is David Wechsler's tests. Scores are set so the population average is **100** and the **standard deviation is 15**, following the bell-shaped **normal curve**.

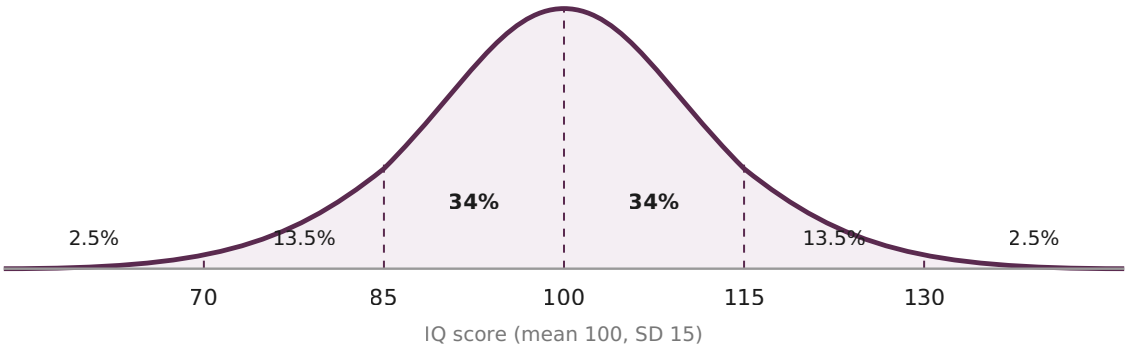


Figure 1. The normal curve for IQ (approximate). About 68% of people score between 85 and 115, about 95% between 70 and 130.

A good test must be **reliable** (gives consistent scores on retesting) and **valid** (measures what it claims to, and predicts what it should). IQ tests are highly reliable. They predict school results well and job performance moderately. A large 2022 re-analysis by Paul Sackett and colleagues revised earlier estimates downward, putting general mental ability among the useful predictors of job performance but behind well-designed structured interviews (Part 11).

Nature, nurture and the Flynn effect

Twin and adoption studies put the **heritability** of adult IQ at roughly 50%, higher in adulthood than childhood. Heritability is a statement about variation within a population in a given environment — not a claim that intelligence is fixed in an individual. The **Flynn effect** proves the point: average scores on IQ tests rose by about three points per decade

across much of the twentieth century, far too fast for genes, pointing to better nutrition, schooling and more abstract everyday thinking. In some countries the rise has recently stalled or reversed.

Emotional intelligence

Emotional intelligence (EI) — perceiving, understanding and managing emotions in yourself and others — became famous through Daniel Goleman's 1995 book, which suggested it could matter more than IQ. Research supports EI as a real and somewhat useful construct, especially in jobs heavy on emotional labour such as sales and service. But much of what it predicts overlaps with personality and general ability, and the "matters more than IQ" claim is not supported.

WORKED EXAMPLE 2 · READING A CUT-OFF

A recruiter screens graduates with an aptitude test scaled like IQ (mean 100, SD 15) and sets the cut-off at 130. What share of the population passes?

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130 = 100 + 2 × 15  → two SDs above the mean
Above +2 SD on a normal curve ≈ 2.5%
(precise value ≈ 2.3%)
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About 1 in 40. If the applicant pool is a typical cross-section, a cut-off that high rejects nearly everyone — including many who would perform well, since test scores predict performance only moderately. Cut-offs should be set from evidence about the job, not from what sounds rigorous.

6 Language

Language is a system for combining symbols into meaningful messages. Its layers: **phonemes** (basic sounds), **morphemes** (smallest units of meaning — "unpaid" has two), **grammar** and **syntax** (rules for combining words), **semantics** (meaning), and **pragmatics** (how meaning depends

on social context — "Can you send the order?" is a request, not a question about ability). Pragmatics is where most cross-regional and cross-language misunderstandings in a national sales force happen.

Does language shape thought? The strong version of the **Sapir–Whorf hypothesis** — that language determines what you can think — has been rejected. A weaker version has some support: Russian has separate basic words for light and dark blue, and Russian speakers distinguish those shades slightly faster. Language nudges attention and memory; it doesn't imprison thought. The business equivalent: the words a company uses for things ("dealer" or "partner", "cost" or "investment") shape what people notice.

7 Held up?

Claim	Verdict	Why
Anchoring and framing effects	**Robust**	Replicated in large multi-lab projects, though effect sizes vary.
Losses loom about twice as large as gains	**Mostly**	Strong for meaningful stakes; some researchers argue it's weaker or absent for small amounts. The "2×" is an average, not a constant.
Reading words about old age makes you walk slower	**Failed**	A famous 1996 "priming" study did not replicate in 2012; many similar behavioural-priming effects have also failed.
There are many independent intelligences	**Weak**	The proposed intelligences correlate; little evidence for independence.
Group brainstorming produces more ideas	**No**	Individuals working separately, then pooling, consistently do better.
Emotional intelligence matters more than IQ	**Overstated**	EI adds some prediction in emotion-heavy jobs; not more than general ability.

Summary

- Thinking uses **mental images** and **concepts**; natural concepts are judged against a **prototype**.
- Problems are solved by trial and error, **algorithms** or **heuristics**. Barriers: functional fixedness, mental set, and **confirmation bias**.
- Judgement is mostly fast **System 1**. Predictable biases: **availability**, **representativeness** with base-rate neglect, **anchoring**, **framing**, **loss aversion**, sunk cost and overconfidence.
- When base rates are low, even good screening produces mostly false positives — a "star" rating meant only about a 31% chance in the worked example.
- Silent individual idea generation beats group brainstorming.
- Intelligence shows a strong general factor (**g**). IQ is scaled to mean 100, SD 15; tests are reliable and moderately predict job performance. Multiple-intelligence theory has weak support.
- Adult IQ is about 50% heritable, yet scores rose about three points per decade in the twentieth century: heritable doesn't mean fixed.
- **Emotional intelligence** is useful in emotion-heavy jobs but not more important than general ability.
- Language has layers from phonemes to pragmatics. It nudges thought but doesn't determine it.

Practice problems

P1. Redo worked example 1 with a better interview: it rates 80% of future stars as "star" but only 5% of others. What is P(top performer | rated star)?

P2. Name the bias: (a) after a flood story in the news, a manager overestimates monsoon losses; (b) a brand keeps funding a failing sub-brand "because of the investment"; (c) a buyer's counter-offer stays close to your opening price.

P3. Rewrite this message in a loss frame: "Dealers who pay within 15 days earn a 2% early-payment discount."

P4. Using the normal curve, what share of people score between 85 and 130?

P5. Your team plans a one-hour group brainstorm for festive-season promotion ideas. Redesign the session using the evidence.

P6. A training vendor says its programme raises "multiple intelligences" and EI "which matter more than IQ". Write three questions to ask it.

Glossary

Prototype

The most typical example of a concept.

Algorithm / heuristic

A step-by-step procedure that guarantees an answer / a fast rule of thumb.

Functional fixedness

Seeing objects only in their usual function.

Confirmation bias

Favouring evidence that supports existing beliefs.

System 1 / System 2

Fast, intuitive thinking / slow, deliberate thinking.

Availability heuristic

Judging likelihood by how easily examples come to mind.

Representativeness heuristic

Judging category membership by resemblance to a prototype.

Base rate

How common something is in the population.

Anchoring

Estimates pulled toward an initial number.

Framing

Different decisions from the same facts described differently.

Loss aversion

Losses weighing more than equal gains.

Sunk-cost fallacy

Letting past, unrecoverable costs drive future decisions.

Divergent / convergent thinking

Generating many answers / narrowing to one.

g (general intelligence)

A general mental ability underlying performance across tests.

Reliability / validity

Consistency of a measure / whether it measures what it claims.

Heritability

The share of variation in a trait within a population linked to genetic differences.

Flynn effect

The rise in average IQ scores across generations.

Pragmatics

How social context shapes the meaning of language.

Answer key

P1. Stars rated star: 80% × 10 = 8. Others rated star: 5% × 90 = 4.5. P = 8 / 12.5 = **64%**. Fewer false alarms doubles the value of the rating.

P2. (a) availability; (b) sunk-cost fallacy; (c) anchoring.

P3. For example: "Dealers who pay after 15 days lose their 2% discount" — or show the invoice at the discounted price and the late-payment price as an added charge.

P4. 85 to 100 = 34%; 100 to 115 = 34%; 115 to 130 = 13.5%. Total ≈ **81.5%**.

P5. Ten minutes of silent, individual idea writing; ideas posted anonymously; the group then clusters, builds on and scores them; the final shortlist is chosen by a pre-agreed criterion, not by whoever spoke loudest.

P6. For example: What independent evidence shows these "intelligences" are separate abilities? What measured change in job performance has the programme produced, against a control group? How does the EI measure relate to existing personality and ability tests?

Sources: chapter structure follows Ciccarelli & White, *Psychology*, 4th ed. (Pearson), Ch. 7; text and figures are original. Tversky & Kahneman, *Science*, 1974 and 1981. Kahneman, *Thinking, Fast and Slow*, 2011. Brainstorming: Diehl & Stroebe, 1987. Priming replication: Doyen et al., *PLOS ONE*, 2012. Selection validity: Sackett et al., *Journal of Applied Psychology*, 2022. Flynn effect: Flynn, 1987 and later reviews. Worked-example figures are illustrative.